Tunable Inter-Body Hole Coupling in a Divided-Body SOI Single-Transistor Latch: Series-STL Equivalence, Continuous Tuning, and a Body Hole-Balance Model

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***Abstract*—A divided-body silicon-on-insulator (SOI) single-transistor latch (STL), in which a resistively grounded n^+ island splits the floating body into two series STLs inside one device, is studied with two-dimensional drift-diffusion TCAD. With hole transport across the island suppressed, the device latches twice, but it then reproduces two separately simulated STLs wired in series within 25 mV. Without lifetime killing, holes cross the island and couple the bodies: the second latch drops from 4.78 to 1.97 V, and the left STL departs from the series-circuit characteristic by up to 1.29 V with negative differential resistance. The coupling is tuned continuously by the island lifetime or, more practically, the island width (15–200 nm). With the first latch fixed at 1.44–1.45 V, the second latch moves between 1.97 and 4.78 V and the second-stage hysteresis between 0 and 2.27 V; a killed-island control rules out a gate-length effect, and at 15 nm the second latch becomes a continuous turn-on. A body hole-balance model built from solution snapshots explains these results. The island hole transmission falls as $\exp(-W_{IS}/L_p)$ with $L_p = 21$ nm and scales with $\sqrt{\tau}$; the impact-ionization multiplication depends on the island potential only, and the inverse current gain on the current only, for every run. Without per-run fitting, the model reproduces the second-latch turn-off of all runs within 0.08 V, the slanted on-branch of the down sweep, and the turn-on and jump height for 30–120 nm islands.**

***Index Terms*—Single-transistor latch (STL), silicon-on-insulator (SOI), floating body, impact ionization, parasitic bipolar transistor, multi-state latch, hole coupling, compact model, carrier lifetime, TCAD.**

I. Introduction

In a floating-body SOI MOSFET, holes generated by impact ionization (II) at the drain junction charge the body, forward-bias the body–source junction and turn on the parasitic lateral n-p-n bipolar transistor, whose electron injection increases II again. When this loop gain exceeds unity the drain current jumps by several decades and returns at a lower voltage: the single-transistor latch (STL) [1]. STLs have been used as bistable resistors (biristors) [2], capacitorless 1T-DRAM [3], [4], leaky integrate-and-fire neurons [5]–[7] and oscillators [8], and low-voltage operation has been pursued [9]. These applications use two states. Ternary logic and multi-level elements need a third state; CMOS-compatible approaches include tunneling-based T-CMOS [12], a threshold switch in series with a MOSFET [10], [11], [13], feedback FETs [14] and gate–body diode MOSFETs [15]. A biristor threshold switch in series with a MOSFET [10], [11] keeps leakage low but needs two devices. Recently a bulk MOSFET with two latches, the triristor, was demonstrated [16]; its two latches share one p-well and are separated only by two current paths, so they can merge with bias.

A natural SOI alternative is to split the floating body physically. An n^+ island across the film divides the body into a left (source–island) and a right (island–drain) STL, and grounding the island through R_{tap} lets the right STL latch first and then raise the island potential until the left STL latches. This paper first shows how such a device reaches two separate latches and why the simpler structures latch only once. It then asks the question a reviewer of any such device should ask: is a fully decoupled divided-body device anything more than two STLs in series? We show that it is not—it reproduces a series STL pair—and that what a single device adds is the *hole coupling* between the two bodies across the island. That coupling makes the device depart from any series circuit, and its strength can be set continuously by the island lifetime or width.

The contributions are: (1) a step-by-step account of why body-doping splits, thickness splits, split gates, floating and tapped islands give one latch, and a hole-balance explanation of the two latches of the decoupled device (Sections III–IV); (2) a direct comparison with two separately simulated STLs in series, and three signatures of the coupled device that a series circuit cannot produce (Section V); (3) continuous tuning of the second latch and of the second-stage hysteresis by two knobs that collapse onto L_p/W_{IS}, with a gate-length control and the coupling limit (Section VI); (4) a body hole-balance model, extracted from solution snapshots, that ties the island width and lifetime to one transmission length and reproduces the second-latch turn-on, turn-off and on-branch slope without per-run fitting (Section VII).

II. Device and Simulation Method

The reference device (Fig. 1) has $L_g = 300$ nm, $T_{si} = 50$ nm, $t_{ox} = 20$ nm, $T_{box} = 200$ nm, n^+ S/D of 10^{20} cm^{-3}, a 30 nm n^+ island (10^{20} cm^{-3}) at mid-channel, left and right p-bodies of 6×10^{17} and 7×10^{17} cm^{-3}, $V_G = -0.2$ V, $R_{tap} = 10$ MΩ and a width of 0.5 μm. The island is connected

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Simulation decks, analysis code and key logs are available at https://github.com/ashleysys04-blip/EE761_DividedBody.

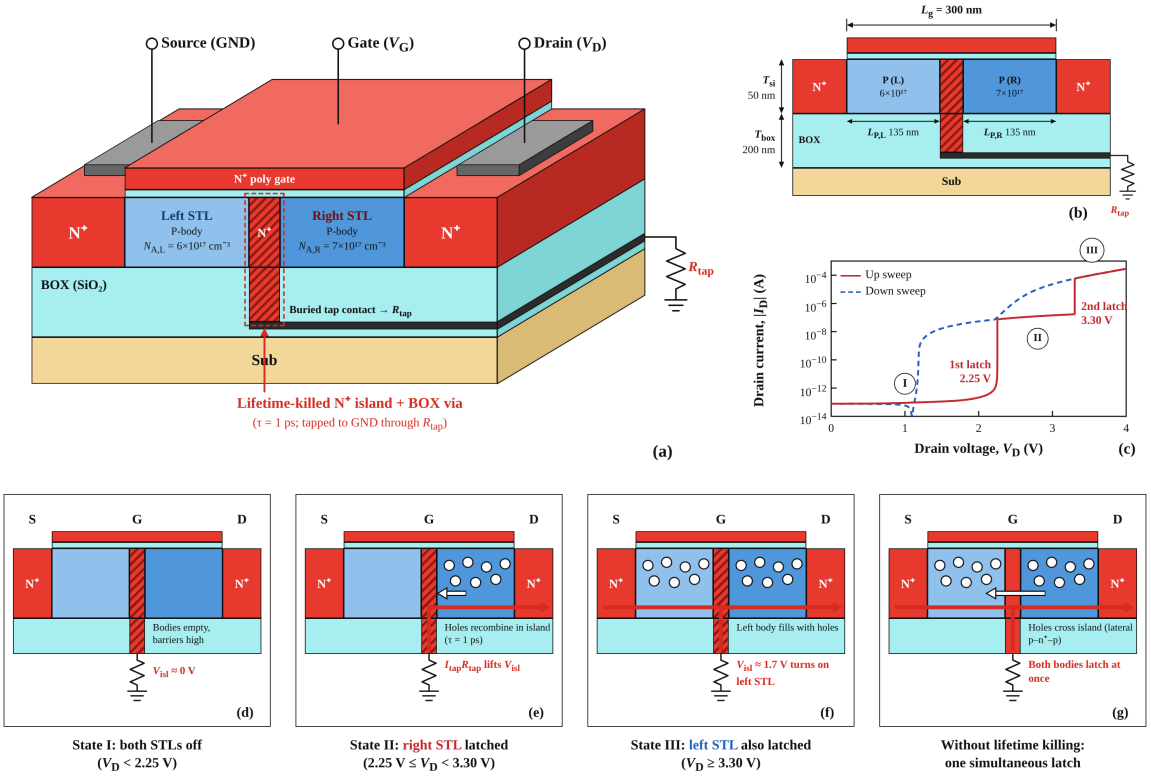


Fig. 1. Proposed divided-body SOI STL. (a) Left and right STLs share one gate; an n^+ island with a BOX via is grounded through R_{tap}. (b) Cross section of the reference device. (c) Two latches of the decoupled reference device (island lifetime 1 ps). (d)–(f) Body holes in states I–III. (g) Without lifetime killing, holes cross the island (lateral p-n^+-p) and both bodies latch at once.

through an n^+ via in the BOX to a bottom tap electrode. Fig. 2 gives the equivalent circuit: each STL is a MOSFET with a parasitic n-p-n transistor on its floating body, charged by an II current source; the island node is the source of the right STL and the drain of the left STL.

Silvaco ATLAS [20] is used with `bgn`, `consrh`, `conmob`, `fldmob`, Fermi statistics, trap-assisted and band-to-band tunneling and the Selberherr II model [18], [19]. With `consrh` the effective SRH lifetime is $\tau_{eff} = \tau_0/(1 + N/N_{SRH})$, $N_{SRH} = 5 \times 10^{16}$ cm^{-3}; lifetimes quoted below are the base value τ_0, so $\tau_{eff} = \tau_0/2001$ in the island and $\approx \tau_0/14$ in the bodies. Because latching produces an S-shaped I–V on which a static voltage sweep stops at the fold, the drain is ramped in time, 0→6→0 V at 0.7 V/ms; the transient crosses the unstable branch and returns the jumps and hysteresis directly. Changing the ramp rate from 0.07 to 7 V/ms moves the first latch by 0.1–0.2 V and the second by less than 0.06 V. The final runs use MixedMode (device plus R_{tap} and sources), which agreed with device-only runs to four digits and was 20 times faster. A latch is defined as a change of the path current by more than one decade at a fixed applied voltage (within 2 mV) that reaches a real current level; numerical artifacts are excluded by checking the time step, the KCL residual, internal potentials and the gate displacement current.

III. Why the Device Latched Only Once

Fig. 3 summarizes the structures tried before two latches were obtained. *Body-doping split.* Small (4–12%) and large

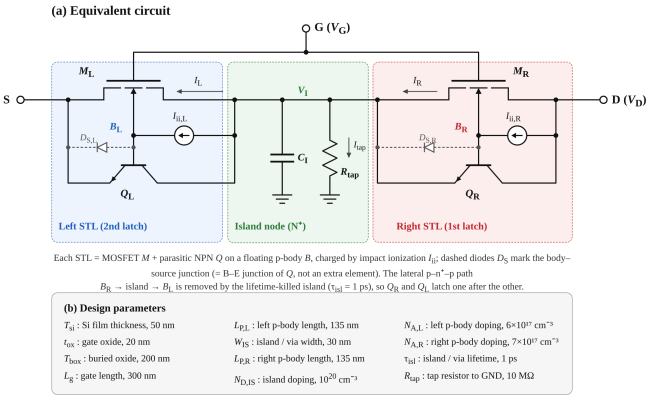


Fig. 2. (a) Equivalent circuit of the divided-body STL. In the decoupled device the island removes the lateral hole path; in the coupled device (Sections V–VI) an additional body-to-body hole current source is needed. (b) Design parameters of the reference device.

(2.3–10 times) left/right doping differences in a single body always gave one latch; the hole quasi-Fermi level stayed continuous across the doping step, because a p/p^+ step does not block majority holes, so the two regions cannot float to different potentials. An oxide notch under the body with a thin Si bridge, a left/right T_{si} split, a split gate and an $n^+/n^-/p/n^+$ stack also gave one latch. *Floating island.* A 30 nm n^+ island separated the two body potentials before latching (−0.02 and 0.36 V), but both bodies and the island jumped together

at 3.26 V [Fig. 3(a)]: with a floating island the right-STL current must flow through the left STL (series constraint). *Tapped island.* Grounding the island through R_{tap} removed this constraint, but at R_{tap} = 600, 60 and 10 kΩ the latch stayed at 2.990 V; zooming into the jump, the right body rose first and the left body followed within ≈ 10 ns [Fig. 3(b)]. Holes injected from the right body into the 30 nm island crossed it and charged the left body: the right p-body, the island and the left p-body form a lateral p-n$^+$-p transistor. A 100 nm island still coupled the bodies, and a top ohmic contact on the island let holes pass underneath it. Only reducing the lifetime of the island and via to 1 ps left the left STL off while the right STL latched alone [Fig. 3(c)]. Raising R_{tap} to 10 MΩ then lifted the island potential enough to drive the left STL, and a standalone left STL showed that a body doping of 6×10^{17} cm^{-3} is needed for an abrupt latch (3 and 5×10^{17} turned on gradually).

IV. Decoupled Device: Two Latches and Their Mechanism

With the island killed (1 ps), the reference device latches at 2.251 V (right STL; I_{tap} jumps from 10^{-12} to 7×10^{-8} A, state I→II) and at 3.300 V (left STL; II→III), when the island potential reaches 1.67 V, close to the latch voltage of the standalone left STL in the same model (1.78 V). On the way down the left and right STLs turn off at ≈ 2.24 and ≈ 1.20 V, so both latches have hysteresis [Fig. 1(c)].

Twenty-eight solution snapshots were analyzed per STL: source–body electron barrier, II hole current $I_{ii} = qW\int G_{ii}dA$ and recombination in the body and in the n$^+$ region that acts as its source [Fig. 4]. For the first latch the right drain-junction field is 1.16×10^6 V/cm ($M - 1 \approx 0.25$); between 2.24 and 2.26 V, I_{ii} jumps by four decades and the right barrier collapses from 0.60 to 0.28 eV. After latching, I_{ii} equals body plus island recombination (e.g., $1.96 \times 10^{-8} \approx 1.06 \times 10^{-8} + 1.00 \times 10^{-8}$ A at 3.0 V). For the second latch the field at the left side of the island rises from 3.6×10^5 to 7.1×10^5 V/cm as the island potential rises, and between 3.28 and 3.32 V the left barrier collapses from 0.61 to 0.10 eV. The G_{ii} map shows generation only at the two drain-side junctions; holes from the right body recombine in the island, so the left STL is turned on only by its own drain field. The turn-off is set by the hole margin: at $\tau_{0,Si} = 10^{-7}$ s only 29% of the generated holes recombine at the last on-point, so the latch survives to low current, whereas at $\tau_{0,Si} = 10^{-8}$ s 78% are lost and the left STL falls by six decades within 20 mV at a holding current of 5.5×10^{-7} A.

Eighty-six single-parameter splits show that the two latches are designed independently: the right-body doping moves the first latch (1.24–2.51 V), the left-body doping the second (2.28–4.01 V), V_G moves both, R_{tap} divides the jump heights between them, and a shorter Si lifetime makes the turn-off abrupt at the cost of higher latch voltages. With $\tau_{0,Si} = 10^{-8}$ s and the island moved to 60% of L_g (left body 165 nm, right body 105 nm), the two hysteresis loops lie at 1.23–1.45 V and 2.56–4.78 V and are separated by 1.12 V. This device (island 30 nm, island lifetime 1 ps) is the decoupled reference for the rest of the paper.

TABLE I
Series-Circuit Control (V; Turn-Off at $|I_S| < 10^{-8}$ A)

	1st on	2nd on	2nd off
one device, metal core	1.499	4.616	2.081
two STLs in series (CUT)	1.476	**4.613**	2.061
two STLs in series (SD)	1.471	4.428	2.019
one device, coupled island	1.444	**1.969**	1.747

V. Decoupled Device Equals a Series STL Pair

To test whether the decoupled device adds anything to a circuit, the island of the decoupled reference was replaced by a metal core through the Si film with a 10 nm n$^+$ shell (an implementable form of island recombination; first and second latch 1.499/4.616 V), and this device was then cut through the middle of the core into two independent ATLAS devices, each keeping its geometry, mesh and half of the core as its internal electrode. The two STLs were wired in series in MixedMode with the same R_{tap} and gate (CUT). A realistic pair of discrete SOI MOSFETs with 500 nm n$^+$ S/D and top contacts (SD) was also simulated. CUT reproduces the one-device result within 3 mV at the second latch and 25 mV elsewhere (Table I, Fig. 5(a)); SD differs by 0.19 V only through its geometry. A divided-body device with no hole transport across the island is therefore exactly a series STL pair: the separate latches alone are not a new function, only a more compact layout.

Keeping the island lifetime at the Si value ($\tau_{0,isl} = 10^{-8}$ s, "coupled") changes this. The second latch drops by 2.6 V to 1.969 V, and the device departs from the series circuit in three ways. First, in a series circuit each STL current is set by its own terminal voltages, so the left-STL operating point ($|I_S|$, V_{tap}) must lie on the I–V of a standalone left STL with the same body (Fig. 5(b), gray). The killed-island device follows that curve within 0.03 V at 1–3 mA (2.269 vs 2.272 V at 1 mA). The coupled device carries the same current at a much lower voltage (1.724 V at 1 mA; 1.519 vs 2.469 V at 3 mA). Second, above 1 mA its left-STL voltage decreases as the current increases, a negative differential resistance that a monotonic standalone on-branch cannot produce. Third, a jump appears at 3.38 V. A hole balance of each body from the snapshots (generation, recombination and $\int J_p\,dy$ at the island edges) shows that the fraction of the left-body hole supply that has crossed the island rises from 23% at 3.0 V to 59% at 3.5 V: the jump closes an n-p-n-p-n hole loop from the right body through the island into the left body. With the island killed this fraction is zero and the jump is absent. Because the jump is only a factor of 2.5 in current and occurs near 6 mA/μm in an isothermal simulation, it is used here only as evidence of coupling, not as a fourth state. In circuit terms (Fig. 2), the coupled device needs a body-to-body hole current source in addition to the R_{tap} node; its magnitude is the coupling strength.

VI. Coupling Strength as a Design Knob

Holes entering the n$^+$ island diffuse as minority carriers over $L_p = \sqrt{D_p\tau_{eff}}$; with $\tau_{eff} = \tau_{0,isl}/2001$ and $D_p \approx 2$ cm^2/s, $L_p \approx 32$ nm$\times\sqrt{\tau_{0,isl}/10^{-8}\text{ s}}$ (the hole transmission

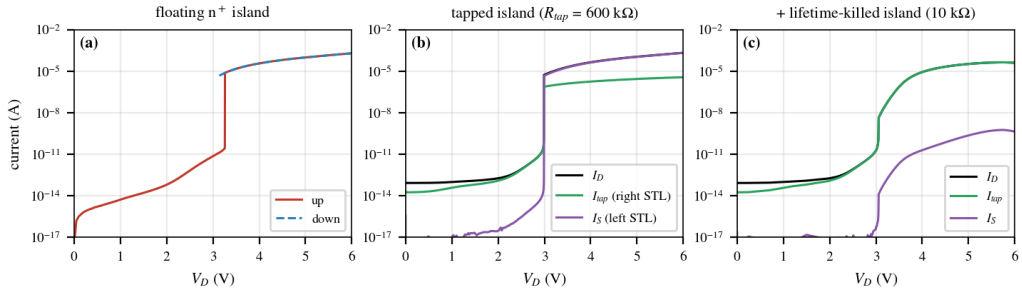


Fig. 3. Latching of the intermediate structures. (a) Floating n^+ island: both STLs latch together at 3.26 V. (b) Island grounded through $R_{tap} = 600$ kΩ: the right (I_{tap}) and left (I_S) STLs jump together at 2.99 V (lateral p-n^+-p coupling). (c) Island lifetime 1 ps: only the right STL latches.

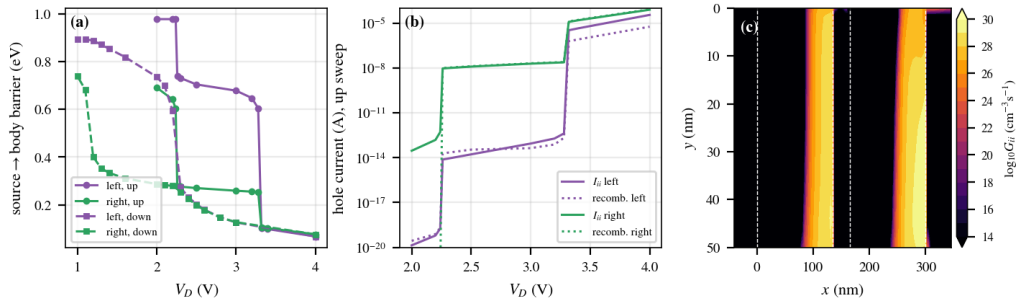


Fig. 4. Snapshot-based mechanism of the decoupled reference device ($\tau_{0,Si} = 10^{-7}$ s). (a) Source-to-body electron barrier of each STL (solid: up, dashed: down). (b) II hole current (solid) and recombination (dashed) on the up sweep. (c) G_{ii} map just after the second latch (3.32 V); dotted lines: source edge, island, drain edge.

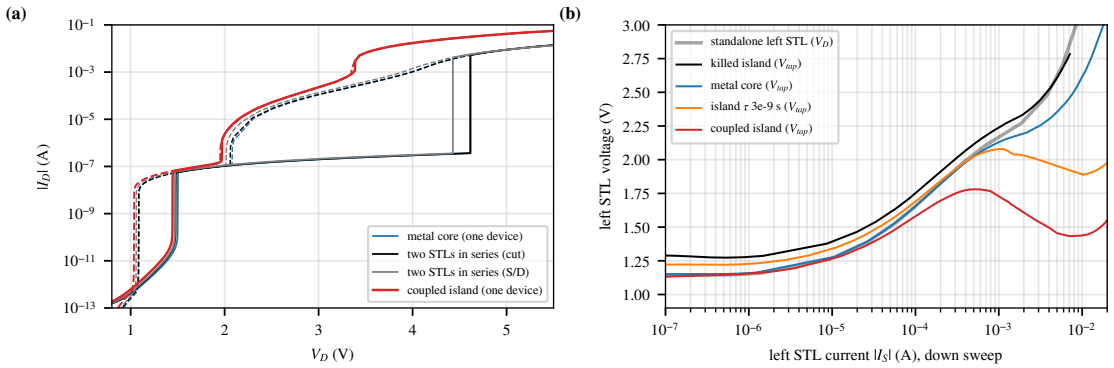


Fig. 5. Series-circuit control. (a) $|I_D|$–V_D (solid: up, dashed: down) of the metal-core device, two separately simulated STLs in series (CUT, SD) and the coupled device; CUT overlaps the metal-core device. (b) Left-STL operating point (V_{tap} vs $|I_S|$, down sweep) against a standalone left STL (V_D vs $|I_D|$, gray). A series circuit must lie on the gray curve.

measured in Section VII gives 21 nm, a uniform rescaling of the L_p/W_{IS} axis). The coupling should therefore scale with L_p/W_{IS}, and it was varied in two ways from the reference: (i) the island lifetime $\tau_{0,isl} = 10^{-12}$–10^{-8} s at $W_{IS} = 30$ nm (only the two lifetime lines of the deck differ), and (ii) the island width $W_{IS} = 15$–200 nm at $\tau_{0,isl} = \tau_{0,Si} = 10^{-8}$ s. The width sweep keeps both body lengths and widens only the island, so $L_g = 270$ nm $+ W_{IS}$ grows with it; a killed island at $W_{IS} = 120$ nm separates this gate-length effect.

The results are listed in Table II and Fig. 6. The first latch (right STL) stays at 1.444–1.449 V in every case; only the second latch moves. In the lifetime sweep it stays at 4.74–4.78 V (series circuit) up to $\tau_{0,isl} = 10^{-10}$ s and falls to 2.99 V at 10^{-9} s and 1.97 V at 10^{-8} s. In the width sweep it falls from 4.31 V (200 nm) through 3.34, 2.58, 2.19 and 2.06 to 1.97 V (30 nm). The two sweeps fall on one curve versus L_p/W_{IS} [Fig. 6(b)]; at equal L_p/W_{IS} they differ by at most ≈ 0.3 V, which we attribute to the one-dimensional diffusion picture behind L_p/W_{IS}. Killing the 120 nm island gives 4.761/2.577 V, the same as the killed 30 nm island (4.779/2.565 V), although L_g grows from 0.30 to 0.39 μm; the width dependence of the coupled devices is therefore entirely due to coupling. The island width, a mask dimension, is the practical knob, whereas the lifetime knob needs a local, extreme lifetime control.

Hysteresis. The second turn-off stays at 2.00–2.04 V for all coupled widths from 45 to 200 nm, so the width moves only the turn-on and the second-stage hysteresis is tuned

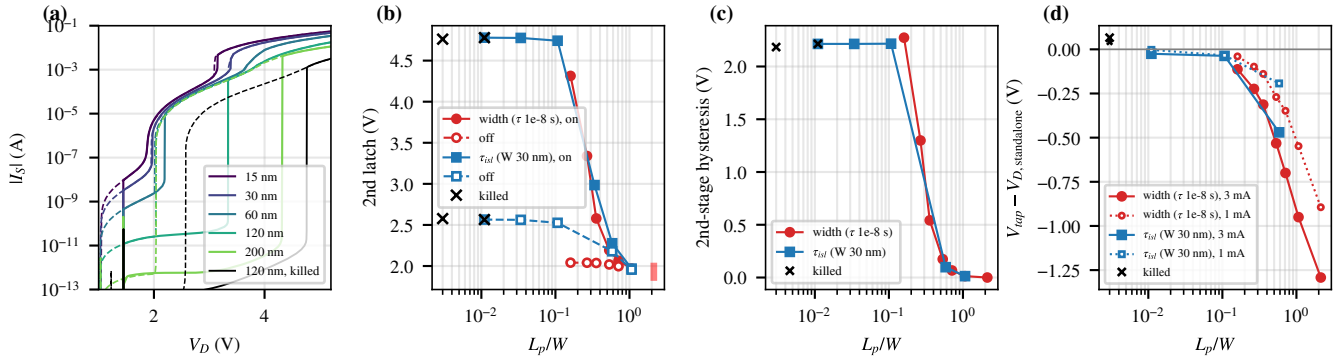


Fig. 6. Continuous tuning of the coupling. (a) Left-STL current of the width sweep (solid: up, dashed: down; black: 120 nm, island killed). (b) Second-latch turn-on (filled) and turn-off (open) versus L_p/W_{IS}: red, width sweep; blue, island-lifetime sweep; ×, killed islands (30, 120 nm); the bar marks the continuous turn-on at 15 nm. (c) Second-stage hysteresis (0 at 15 nm). (d) Deviation of the left-STL operating point from the series circuit (standalone left STL) at $|I_S|$ = 3 mA (solid) and 1 mA (dotted), down sweep.

TABLE II
SECOND LATCH VERSUS COUPLING STRENGTH. ON/OFF: ONE-DECADE JUMP AT FIXED APPLIED VOLTAGE (V); JUMP: DECADES OF $|I_D|$ AT TURN-ON

	L_p/W_{IS}	2nd on	2nd off	hyst.	jump
W_{IS} = 15 nm	2.13	cont.[a]	same curve	0	none
30 nm ($\tau_{0,isl} = 10^{-8}$ s)	1.07	1.969	1.956	0.01	0.8
45 nm	0.71	2.064	1.997	0.07	1.3
60 nm	0.53	2.193	2.019	0.17	1.8
90 nm	0.36	2.579	2.037	0.54	2.5
120 nm	0.27	3.339	2.040	1.30	3.2
200 nm	0.16	4.314	2.041	2.27	4.1
$\tau_{0,isl} = 3 \times 10^{-9}$ s (30 nm)	0.58	2.277	2.179	0.10	1.4
10^{-9} s	0.34	2.985	–[b]	–	–
10^{-10} s	0.11	4.742	2.526	2.22	3.7
10^{-11} s	0.03	4.776	2.562	2.21	3.7
killed (10^{-12} s), 30 nm	0.01	4.779	2.565	2.21	3.7
killed, 120 nm	0.003	4.761	2.577	2.18	3.6

[a] Continuous turn-on between 1.86 and 2.0 V. [b] Run stopped at 5.45 V (time limit).

continuously from 2.27 V to zero [Fig. 6(c)]. The low turn-off is itself a coupling effect: at the same 120 nm geometry the left STL turns off at 2.04 V with a coupled island and at 2.58 V with a killed one. Hole maps at 3.0 V show why. Both devices are inside their hysteresis at 3.0 V, with an empty left body on the way up and a filled one on the way down; the difference is the island. The coupled island holds 10^{16}–10^{17} cm^{-3} holes on the way down and keeps feeding the left body, whereas the killed island stays below 10^4 cm^{-3}. A wide island blocks the coupling before latch-on, but after latch-on the hole flux across it is large and the coupling returns, holding the left STL down to a lower voltage. The right-STL turn-off moves in the same direction (1.035 V coupled vs 1.228 V killed).

Limit of strong coupling. Coupling has a cost: the height of the second jump falls from 4.1 decades (200 nm) through 3.2, 2.5, 1.8 and 1.3 to 0.8 decades (30 nm), and at 15 nm the left STL turns on without a jump between 1.86 and 2.0 V and returns along the same curve ($|I_S| = 5.2 \times 10^{-6}$ A up and down at 2.0 V). The two bodies then behave as one hole reservoir. The useful range for an abrupt second latch therefore starts between 15 and 30 nm; we take 30–120 nm as the tuning range and treat 200 nm, where the island is wider than either body, as the limit toward zero coupling. The hole-loop jump remains at 3.10–3.15 V at 15 nm.

Deviation from the series circuit. With the operating-point test of Fig. 5(b) applied to every run [Fig. 6(d)], the left-STL voltage at 3 mA lies -1.29 to -0.11 V from the standalone characteristic for W_{IS} = 15 to 200 nm and returns to $-0.03/+0.07$ V when the 30/120 nm islands are killed; the lifetime sweep lies on the same curve. The departure from a series STL pair thus holds over the whole tuning range, and its size is set by the coupling strength.

VII. BODY HOLE-BALANCE MODEL

To explain why coupling lowers and softens the second latch and why the down sweep is slanted, the hole balance of the left p body was extracted from every solution snapshot of every coupling run (17 per run, 0–6–0 V). With the body between the source junction and the left island edge as a control volume, the generated holes $I_{ii} = qW\int G_{ii}dA$, the recombination $\int U dA$, the holes back-injected into the n$^+$ source and the holes arriving from the island (both as $W\int J_p\,dy$ on cuts 1 nm inside the body) were integrated, together with the island potential V. The balance closes within 12% with the left STL off and 20% with it on. Writing I for the left-STL electron current $|I_S|$,

$$(M(V)-1)\,I+\eta\,I_D+G_0 = I/\beta(I), \qquad I_D = I+V/R_{tap}, \tag{1}$$

where $M-1 = I_{ii}/I$, $1/\beta$ is the hole loss per electron, η is the island hole current per drain-current electron and $G_0 = 10^{-15}$ A is other generation (the off-state hole currents of the killed runs). Three findings make (1) predictive [Fig. 7(a)–(c)]. First, with the left STL off (state II), $\eta = 0.029\exp(-W_{IS}/L_p)$ with $L_p = 21$ nm over 15–200 nm, and scaling L_p by $\sqrt{\tau}$ predicts 2.3×10^{-3} for $\tau_{0,isl} = 3\times10^{-9}$ s against 2.5×10^{-3} extracted: the width and lifetime knobs are one transmission $\exp(-W_{IS}/L_p)$, which is the origin of the L_p/W_{IS} collapse in Fig. 6(b). In the circuit of Fig. 2 the coupling is a hole current source ηI_D into the left body. Second, $M-1$ below 3 mA depends only on V for all runs, killed or coupled. Third, $1/\beta$ is a single function of I for all

runs: it falls as $I^{-0.28}$ from 0.8 at 10^{-11} A (recombination in the depletion region), has a minimum of 0.05 near 1 μA and rises to 0.35 at 10 mA (high injection). Eight parameters, common to all runs, describe $M-1$ (logistic in V) and $1/\beta$; $\eta(I_D)$ is taken from each run's snapshots without fitting.

For every I, (1) gives one island potential $V(I)$, since its left side rises with V. With the right STL in series, $V_D(I) = V(I) + V_R(I_D)$, where $V_R(I_D)$ is the right-STL on-branch read from the same run ($V_D - V_{tap}$ vs $|I_D|$). $V_D(I)$ is S-shaped: the up sweep follows its rising part and jumps at the local maximum (second latch), and the down sweep drops at the local minimum (turn-off); without a local maximum the turn-on is continuous. Fig. 7(d)–(f) compares this with TCAD. The turn-off is reproduced within 0.01–0.08 V for every run, coupled (2.00–2.07 V) and killed (2.58–2.60 V), so the 0.5 V lower turn-off of a coupled island is the island hole current holding the body on. The slanted down sweep follows from (1) on the on-branch, $M-1 = 1/\beta - \eta I_D/I$:

$$\frac{d\ln I}{dV} = \frac{d\ln(M-1)/dV}{d\ln(1/\beta - \eta I_D/I)/d\ln I}, \tag{2}$$

i.e., the field sensitivity of impact ionization divided by the current roll-off of β: as V falls, $M-1$ falls and the balance can only be met at a lower current, so $\log I$ falls almost linearly with V. For 30–120 nm islands the turn-on is within 0.2 V and the jump height rises with the island width as in TCAD (equal at 90–120 nm): the island holes pre-charge the off body and shrink the fold of the S-curve, which is why stronger coupling gives a softer second latch. Two features are not reproduced. The model keeps a 1.5-decade jump at 15 nm, where TCAD turns on continuously, and loses its fold only at five times the 30 nm coupling. The turn-on of the 200 nm and killed islands occurs at island potentials above 3 V, beyond the extracted $M-1$. Both point to the low-injection β below 10^{-11} A and to the current dependence of $M-1$ at high current, which (1) neglects.

VIII. Discussion

Implementation. The two ends of the coupling range have different physical forms. Zero coupling needs an island effective lifetime of $\leq 5\times10^{-14}$ s, which damage alone is unlikely to provide; a metal or silicide core through the full Si thickness gives the same result (Section V; with $\tau_{0,Si} = 10^{-8}$ s and the island at 60%), but it was modeled as an ideal ohmic boundary with infinite surface recombination. Intermediate coupling is obtained simply by the island width at the Si lifetime, so no local lifetime control of the island is needed. The body lifetime of $\tau_{0,Si} = 10^{-8}$ s used for abrupt turn-off is within conventional lifetime control (Au/Pt diffusion, electron or proton irradiation) [17], and R_{tap} = 1–10 MΩ can be a high-resistance poly-Si resistor.

Use of the coupling knob. With the first latch fixed, the coupling sets where the second latch fires and how wide its hysteresis is. A wide second-stage window (weak coupling) suits a three-state memory or a ternary threshold element; a narrow window (strong coupling) gives a second, nearly reversible switching step. The decoupled device behaves as a series pair, so circuit models of series STLs, including current-driven relaxation oscillation, apply directly; for the coupled device the island node needs the additional hole current source ηI_D of (1), whose width dependence is $\exp(-W_{IS}/L_p)$.

Limitations. All results are 2D, isothermal drift-diffusion. Local-field II overestimates generation in short channels: in the standalone left STL the energy-balance (hydrodynamic) latch was 0.7–0.8 V higher than the DD one. Full-device energy-balance runs of the decoupled device stopped at 3.6–5.3 V because the electron temperature at the right drain junction exceeded 10^4 K and the time step collapsed, despite coupled Newton, a 1 nm mesh, relaxed tolerances, MixedMode and graded drains. An energy-balance run of the 30 nm coupled device, whose second latch lies lower, shows weaker II (no first latch below 3 V) and had not converged beyond 3.06 V; the second latch is therefore a DD result. An attempt to calibrate the II coefficients to a measured double-latch bulk device [16] matched its Id–Vg but not its first latch, and a 20% change of the Selberherr B coefficient moved a coefficient-sensitive latch by $\approx$20%. Absolute voltages are thus uncertain by roughly $\pm$20%, whereas the comparisons in this paper (coupled vs killed vs series circuit at the same model) do not depend on the coefficients. Self-heating at the mA/μm currents of state III is not included.

IX. Conclusion

A divided-body SOI STL with a resistively grounded n^+ island latches twice once hole transport across the island is removed, and each latch is the point where II hole generation in one body overcomes recombination. Such a decoupled device is, however, a series STL pair: two separately simulated STLs in series reproduce it within 25 mV. What a single device adds is the hole coupling between the bodies. The coupled device departs from the series characteristic by up to 1.29 V at 3 mA, shows negative differential resistance and a hole-loop jump, and its coupling is tuned continuously by the island lifetime or, more practically, the island width; both collapse onto L_p/W_{IS}. With the first latch fixed at 1.44–1.45 V, the second latch spans 1.97–4.78 V and the second-stage hysteresis 0–2.27 V, without a gate-length effect. The strongest coupling (15 nm) removes the second latch and sets the lower bound of the useful width. A body hole-balance model with an island hole transmission $\exp(-W_{IS}/L_p)$, $L_p = 21$ nm, and universal $M-1(V)$ and $\beta(I)$ reproduces the second-latch turn-off of all runs and the turn-on of 30–120 nm islands without per-run fitting, and explains the slanted down sweep as the field sensitivity of impact ionization over the current roll-off of β. Remaining work is the low-injection β that sets the coupling limit, energy-balance confirmation of the second latch and calibration against measured SOI devices.

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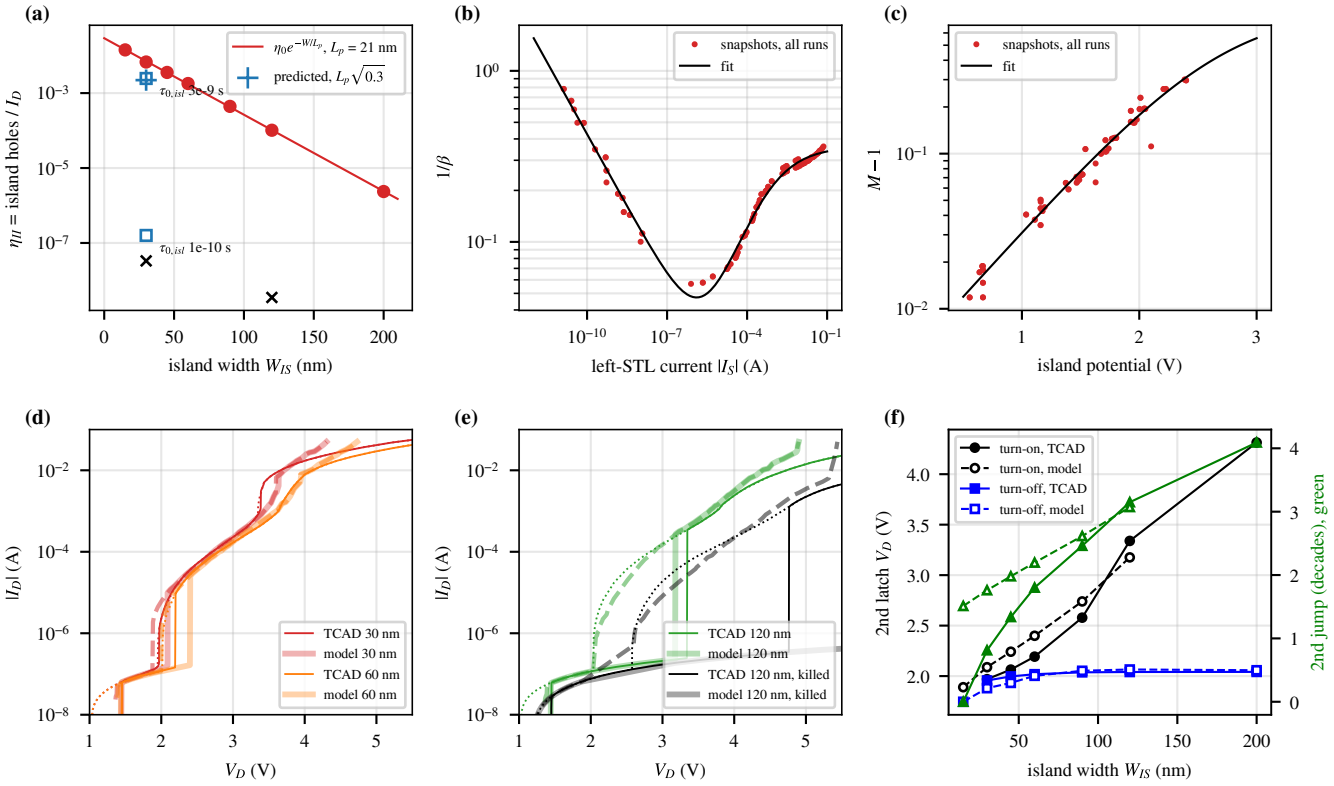


Fig. 7. Body hole-balance model. (a) Island hole current per drain current with the left STL off, versus island width (circles), island lifetime (squares; cross: predicted with $L_p\sqrt{\tau/10^{-8}}$ s) and killed islands (×). (b) Inverse current gain $1/\beta$ and (c) $M-1$ of the left STL from the snapshots of all runs, with the common fits. (d), (e) $|I_D|$–V_D from TCAD (thin; solid up, dotted down) and from (1) with the right-STL on-branch in series (thick; solid up, dashed down). (f) Second-latch turn-on, turn-off and jump height versus island width, TCAD (filled) and model (open).

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